Scaling language models

He He

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Logistics

- We will drop the lowest quiz.
- Solutions will be put up on campuswire.
- Plan for today:
  - LM basics for 30min (He)
  - Scaling laws (Nick)

Recap: pretraining and finetuning

Key idea:

- Pretraining on large textual data to learn many tasks/capabilities
- Finetuning to adapt the model to downstream tasks

Challenges in large scale pretraining:

- Tokenize multilingual and multimodal data
- Efficient training over long sequences
- Gradient vanishing/exploding, numerical stability

What solutions have we studied?

What do language models do?

- Answer questions
- Summarize documents
- Write programs
- Prove theorems
- Write jokes
- ...

Dial back twenty years

Which sequence is more likely to be an English sentence?

- Speech recognition

  the *tail* of a dog
  the *tale* of a dog

  It's not easy to *wreck a nice beach*.
  It's not easy to *recognize speech*.
  It's not easy to *wreck an ice beach*.

- Machine translation

  He sat on the *table*.
  He sat on the *figure*.

  Such a Europe would *the rejection of any* ethnic nationalism.
  Such a Europe would *mark the refusal of all* ethnic nationalism.

Problem formulation

Goal: Assign probabilities to a sequence of tokens, e.g.,

- p(the red fox jumped) p(the green fox jumped)

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- The set of all sentences (of varying lengths): $\mathcal{V}^*$
- Assign a probability $p(x)$ to all sentences $x \in \mathcal{V}^*$.

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$$p_s(x) = \frac{\text{count}(x)}{N} .$$

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  - Most sentences only occur once. *sparsity issue*
  - Need to restrict the model.

Simplification 1: sentence to tokens

Solve a smaller problem: model probability of each token

Decompose the joint probability using the **probability chain rule**:

$$\begin{aligned} p(x) &= p(x_1, \ldots, x_n) \\ &= p(x_1)p(x_2 \mid x_1)p(x_3 \mid x_1, x_2) \ldots p(x_n \mid x_1, \ldots, x_{n-1}) \end{aligned}$$

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(Doesn't have to go from left to right)

$$= p(x_n)p(x_{n-1} \mid x_n) \ldots p(x_1 \mid x_2, \ldots, x_n)$$

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- Problem reduced to modeling conditional token probabilities
  the red fox $\rightarrow$ jumped
- But there is still a large number of contexts!

Simplification 2: limited context

Reduce dependence on context by the **Markov assumption**:

- First-order Markov model

$$p(x_i \mid x_1, \dots, x_{i-1}) = p(x_i \mid x_{i-1})$$
$$p(x) = \prod_{i=1}^{n} p(x_i \mid x_{i-1})$$

- Number of contexts?

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- Number of parameters? $|\mathcal{V}|^2$

Model sequences of variable lengths

Assume each sequence starts with a special **start symbol**: $x_0 = *$.

Assume that all sequences end with a **stop symbol** $\texttt{STOP}$, e.g.

$$
\begin{aligned}
& p(\text{the}, \text{fox}, \text{jumped}, \texttt{STOP}) \\
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What if we don't have the stop symbol?

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What if we don't have the stop symbol?

- Which one is larger: $p(\text{the fox})$ or $p(\text{the fox jumped})$?

N-gram LM

- Unigram language model (no context):

$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} p(x_i) \ .$$

- Bigram language model ($x_0 = *$):

$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} p(x_i \mid x_{i-1}) \ .$$

- n-gram language model:

$$p(x_1, \ldots, x_m) = \prod_{i=1}^{m} p(x_i \mid \underbrace{x_{i-n+1}, \ldots, x_{i-1}}_{\text{previous } n-1 \text{ words}}) \ .$$

Parameter estimation

Maximum likelihood estimation over a corpus (a set of sentences):

- Unigram LM

$$p_{\mathrm{MLE}}(x) = \frac{\mathrm{count}(w)}{\sum_{w \in \mathcal{V}} \mathrm{count}(w)}$$

- Bigram LM

$$p_{\mathrm{MLE}}(w \mid w') = \frac{\mathrm{count}(w, w')}{\sum_{w \in \mathcal{V}} \mathrm{count}(w, w')}$$

- In general, for n-gram LM,

$$p_{\mathrm{MLE}}(w \mid c) = \frac{\mathrm{count}(w, c)}{\sum_{w \in \mathcal{V}} \mathrm{count}(w, c)}$$

where $c \in \mathcal{V}^{n-1}$.

Example

- Training corpus (after tokenization)

  $$\{\text{The fox is red}, \text{The red fox jumped}, \text{I saw a red fox}\}$$

- Collect counts

  $$\begin{aligned}&\text{count}(\text{fox}) = 3\\&\text{count}(\text{red}) = 3\\&\text{count}(\text{red}, \text{fox}) = 2\\&\ldots\end{aligned}$$

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- What is the probability of “The fox saw I jumped”? Zero probability on unseen n-grams

Generating text from an n-gram model

1. Initial condition: context $= *$
2. Iterate until next_word is $\texttt{STOP}$:
   2.1 next_word $\sim p(\cdot \mid \text{context}[: -(n-1)])$
   2.2 context $\leftarrow$ context $+$ next_word

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1 gram	–To him swallowed confess hear both. Which. Of save on trail for are ay device and rote life have –Hill he late speaks; or! a more to leg less first you enter
2 gram	–Why dost stand forth thy canopy, forsooth; he is this palpable hit the King Henry. Live king. Follow. –What means, sir. I confess she? then all sorts, he is trim, captain.
3 gram	–Fly, and will rid me these news of price. Therefore the sadness of parting, as they say, 'tis done. –This shall forbid it should be branded, if renown made it empty.
4 gram	–King Henry. What! I will go seek the traitor Gloucester. Exeunt some of the watch. A great banquet serv'd in; –It cannot be but so.

What is the training data?

How to evaluate language models?

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Held-out likelihood on test data D (negative test loss):

$$\ell(D) = \sum_{x \in \mathcal{D}} \sum_{i=1}^{|x|} \log p_\theta(x_i \mid x_{1:i-1}) \ ,$$

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- What is the minimum and maximum PPL you can get?

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N-gram language models:

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- MLE estimate: counting n-grams in the training corpus

Evaluation by **held-out perplexity**: how much probability mass does the model assign to unseen text

Challenges:

- **Generalization**: sentences containing unseen n-grams have zero probability
- Much research in n-gram LM is dedicated to **smoothing** methods that allocate probability mass to unseen n-grams

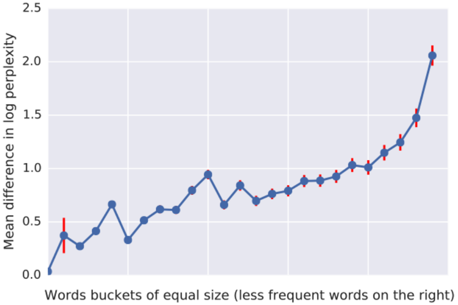
Early efforts on scaling neural language models

Model	Test Perplexity	Number of Params [billions]
Sigmoid-RNN-2048 (Ji et al., 2015a)	68.3	4.1
Interpolated KN 5-gram, 1.1B n-grams (Chelba et al., 2013)	67.6	1.76
Sparse Non-Negative Matrix LM (Shazeer et al., 2015)	52.9	33
RNN-1024 + MaxEnt 9-gram features (Chelba et al., 2013)	51.3	20
LSTM-512-512	54.1	0.82
LSTM-1024-512	48.2	0.82
LSTM-2048-512	43.7	0.83
LSTM-8192-2048 (No Dropout)	37.9	3.3
LSTM-8192-2048 (50% Dropout)	32.2	3.3
2-Layer LSTM-8192-1024 (BIG LSTM)	30.6	1.8
BIG LSTM+CNN Inputs	**30.0**	**1.04**
BIG LSTM+CNN Inputs + CNN Softmax	39.8	**0.29**
BIG LSTM+CNN Inputs + CNN Softmax + 128-dim correction	35.8	**0.39**
BIG LSTM+CNN Inputs + Char LSTM predictions	47.9	**0.23**

Figure: From Exploring the Limits of Language Modeling

Significant improvement in held-out perplexity given similar model sizes ($\sim$1B)

Improvement from neural language models



< S > With even more new technologies coming onto the market quickly during the past three years , an increasing number of companies now must tackle the ever-changing and ever-changing environmental challenges online . < S > Check back for updates on this breaking news story . < S > About 800 people gathered at Hever Castle on Long Beach from noon to 2pm , three to four times that of the funeral cortège . < S > We are aware of written instructions from the copyright holder not to , in any way , mention Rosenberg 's negative comments if they are relevant as indicated in the documents , " eBay said in a statement . < S > It is now known that coffee and cacao products can do no harm on the body . < S > Yuri Zhirkov was in attendance at the Stamford Bridge at the start of the second half but neither Drogba nor Malouda was able to push on through the Barcelona defence .

Figure: From Exploring the Limits of Language Modeling

LSTM vs KN5: improved perplexity on tail words

Recap: language modeling as pretraining

What can we do with a very large language model?

- The cats that are raised by my sister ________ sleeping. *syntax*
- Jane is happy that John invited ________ friends to his birthday party. *coreference*
- ________ is the capital of Tanzania. *knowledge*
- The boy is ________ because he lost his keys. *commonsense*
- John took 100 bucks to Vegas. He won 50 and then lost 100. Now he only has ________ to go home. *numerical reasoning*

Predicting the next word entails many natural language understanding tasks

Recap: Zero-shot behaviors from GPT

Key insight: if the model has learned to understand language through predicting next words, it should be able to perform these tasks *without finetuning*

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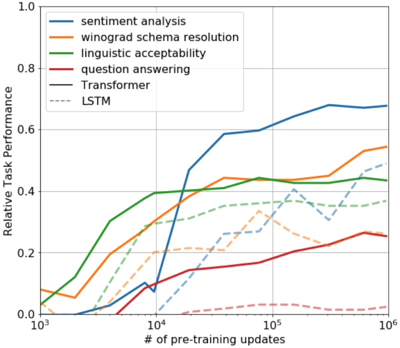


Heuristics for zero-shot prediction:

- Sentiment classification: [example] + very + {positive, negative} *prompting*
- Linguistic acceptability: thresholding on log probabilities
- Multiple choice: predicting the answer with the highest log probabilities

Learning dynamics: zero-shot performance increases during pretraining

GPT-2 details

- Similar to GPT-1 but scaled up (1.5B parameters)
- Data (WebText): $\sim$40GB of web pages scraped from the internet that was curated to include high-quality text
- Tokenization: BPE over byte sequences for universal text processing.
  - Small base vocabulary (256)
  - Can process any text data regardless of pre-processing, tokenization, or vocab size.
- Larger context size (1024 tokens)

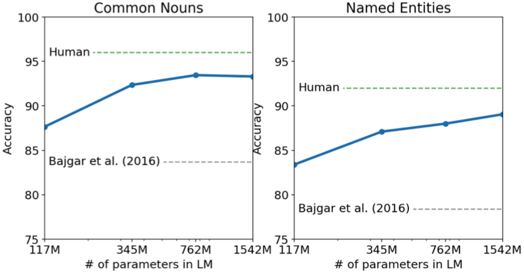
Zero-shot performance: cloze test

S: 1 Mr. Cropper was opposed to our hiring you .
2 Not , of course , that he had any personal objection to you , but he is set against female teachers , and when a Cropper is set there is nothing on earth can change him .
3 He says female teachers ca n't keep order .
4 He 's started in with a spite at you on general principles , and the boys know it .
5 They know he 'll back them up in secret , no matter what they do , just to prove his opinions .
6 Cropper is sly and slippery , and it is hard to corner him . ''
7 `` Are the boys big ? ''
8 queried Esther anxiously .
9 `` Yes .
10 Thirteen and fourteen and big for their age .
11 You ca n't whip 'em -- that is the trouble .
12 A man might , but they 'd twist you around their fingers .
13 You 'll have your hands full , I 'm afraid .
14 But maybe they 'll behave all right after all . ''
15 Mr. Baxter privately had no hope that they would , but Esther hoped for the best.
16 She could not believe that Mr. Cropper would carry his prejudices into a personal application .
17 This conviction was strengthened when he overtook her walking from school the next day and drove her home .
18 He was a big , handsome man with a very suave , polite manner .
19 He asked interestedly about her school and her work , hoped she was getting on well , and said he had two young rascals of his own to send soon .
20 Esther felt relieved .

q: She thought that Mr. ________ had exaggerated matters a little .

C: Baxter, Cropper, Esther, course, fingers, manner, objection, opinion, right, spite.

a: Baxter



Larger models quickly closes the gap with human performance

Zero-shot performance: generative QA

Question	Generated Answer	Correct	Probability
Who wrote the book the origin of species?	Charles Darwin	✓	83.4%
Who is the founder of the ubuntu project?	Mark Shuttleworth	✓	82.0%
Who is the quarterback for the green bay packers?	Aaron Rodgers	✓	81.1%
Panda is a national animal of which country?	China	✓	76.8%
Who came up with the theory of relativity?	Albert Einstein	✓	76.4%
When was the first star wars film released?	1977	✓	71.4%
What is the most common blood type in sweden?	A	✗	70.6%
Who is regarded as the founder of psychoanalysis?	Sigmund Freud	✓	69.3%
Who took the first steps on the moon in 1969?	Neil Armstrong	✓	66.8%
Who is the largest supermarket chain in the uk?	Tesco	✓	65.3%
What is the meaning of shalom in english?	peace	✓	64.0%
Who was the author of the art of war?	Sun Tzu	✓	59.6%
Largest state in the us by land mass?	California	✗	59.2%
Green algae is an example of which type of reproduction?	parthenogenesis	✗	56.5%
Vikram samvat calender is official in which country?	India	✓	55.6%
Who is mostly responsible for writing the declaration of independence?	Thomas Jefferson	✓	53.3%
What us state forms the western boundary of montana?	Montana	✗	52.3%
Who plays ser davos in game of thrones?	Peter Dinklage	✗	52.1%
Who appoints the chair of the federal reserve system?	Janet Yellen	✗	51.5%
State the process that divides one nucleus into two genetically identical nuclei?	mitosis	✓	50.7%

Zero-shot performance: machine translation

- Induce the task through a demonstration example:

$$\text{translation} \sim p(\cdot \mid \text{[french sentence] = [english sentence]; [french sentence] =})$$

- WMT-14 French-English test set: 11.5 BLEU (worse than unsupervised MT)
- But, there's only 10MB french data in the 40GB training data!
    - Typical unsupervised MT methods require crosslingual embeddings or monolingual corpora

GPT-3: scaling up

- GPT-2 shows promise for zero-shot learning, but performance is still unsatisfying
- GPT-3 scales up model size, data size and diversity, and number of training steps
- Notable improvement in zero-shot and few-shot performance
- Inducing a task through natural language task descriptions